Booth Quick Build

The short version of the booth manual: one phone, one Pi, one screen, one printer, in nine steps. About 3 hours for the first booth and 1.5 hours for each one after. When something needs more explanation, the full manual has the why.



What you're building: lift the handset, say a prompt, wait on hold, watch the video, press 0 for a receipt.

What you need for one booth

- [ ] Wall phone (MS-303 beige)
- [ ] CanaKit Raspberry Pi 5 Starter Kit PRO (Pi, power supply, case, memory card, cables)
- [ ] ARZOPA Z1FC 16.1" screen, plus a USB-C charger for it
- [ ] Micro-HDMI to mini-HDMI cable
- [ ] Raspberry Pi Pico W, plus female-to-female jumper wires
- [ ] Sabrent USB audio adapter and 2 × 3.5 mm pigtail cables
- [ ] Rongta 80 mm receipt printer and paper

- [ ] 26 AWG wire in several colours, heat-shrink, zip ties with adhesive mounts, 1" (25 mm) clear heat-shrink for the Pico
- [ ] Power strip

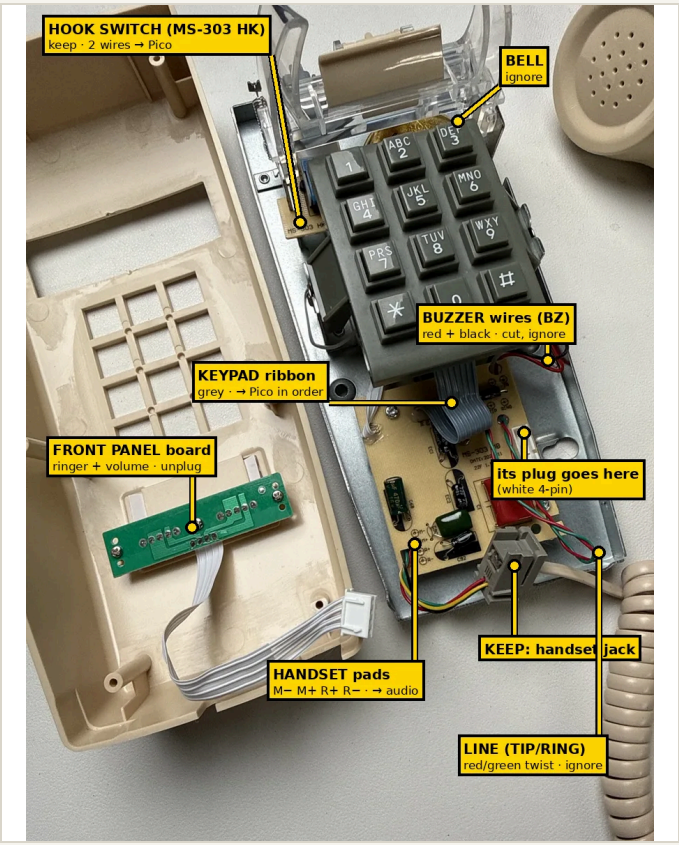
Tools: soldering iron set to 350 °C (660 °F), solder, multimeter, wire strippers, flush cutters, heat gun, small screwdriver, tape and a marker.

STEP 1

Open the phone

15 minutes

1. Unscrew the metal back plate and lift the front shell away gently.
2. Unplug the white plug from the front panel (ringer and volume sliders). It's the only plug; keep its 4 wires for the volume control. Every other wire gets cut.
3. Take a photo before you cut anything. Always cut wires **close to the main board**, so you keep the most length.



What's what. Keep the handset, hook switch and keypad; ignore the bell, buzzer and phone line.

STEP 2

Handset to audio adapter

20 minutes

Cut the four handset wires at the board's M–, M+, R+, R– pads. On the MS-303, **black and yellow are the earpiece** and **red and green are the mic**.

HANDSET WIRE	PAD	PIGTAIL WIRE	PLUG PART
Earpiece pigtail (🎧 jack)			
Black	R+	Red	tip
Yellow	R–	Black	sleeve
Mic pigtail (🎤 jack)			
Red	M+	Red	tip
Green	M–	Black	sleeve

Each pigtail's white wire (the ring) isn't used: snip it short and cover it with heat-shrink. Heat-shrink each joint, then a bigger piece of ½" (13 mm) heat-shrink over each finished splice.

CHECK

Before soldering: black to yellow reads 100–300 Ω on the meter. After: record and play back through the handset. Mic very quiet? Swap red and green.

STEP 3

Hook switch

10 minutes

1. The hook switch board (printed MS-303 HK) has **two white wires** running to **HK** on the main board. Cut both close to the main board and strip 3 mm (⅛ in).
2. Meter on beep, probes on the two wires. Press the plunger, then let go. Note whether it beeps **pressed** or **released**.

CHECK

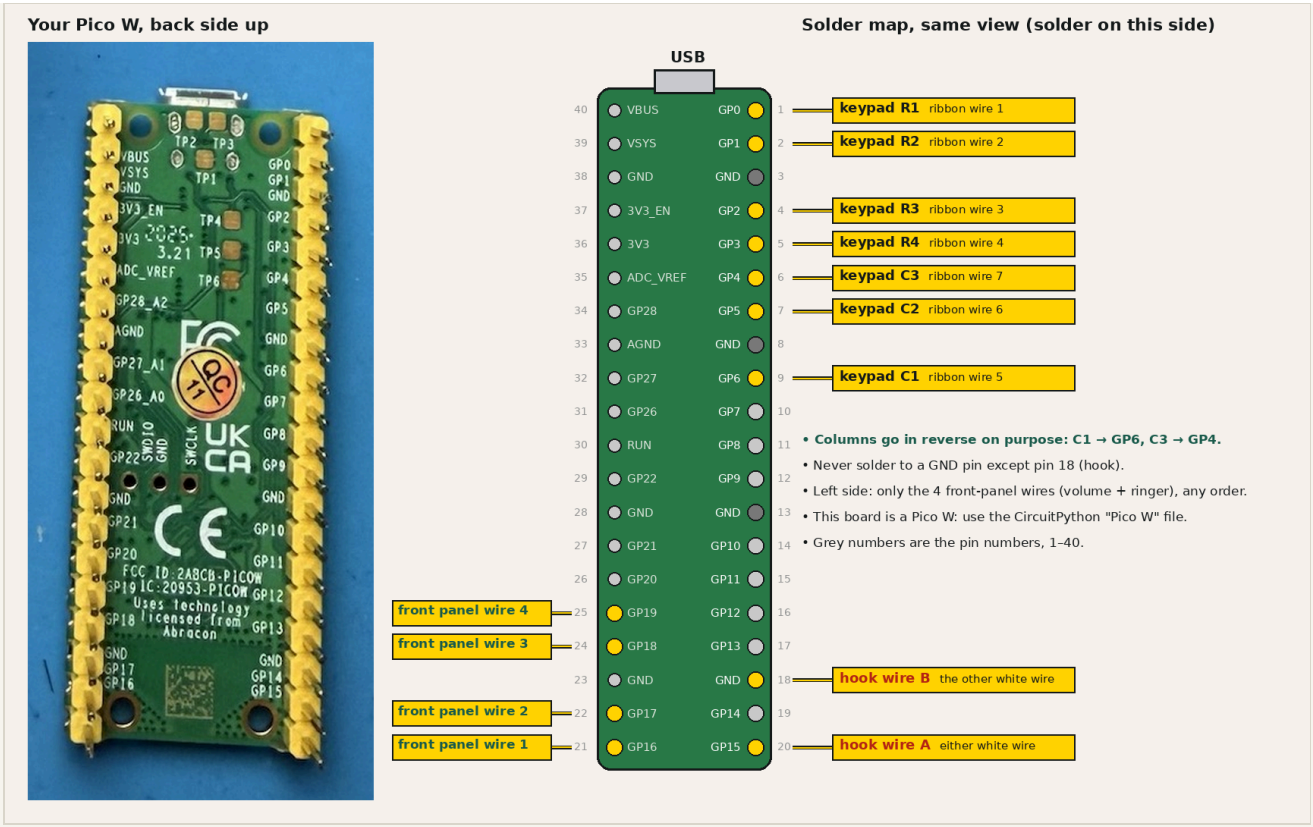
Beeps when **released** (handset lifted): nothing to change. Beeps when **pressed**: in the Pico's `code.py`, set `OFF_HOOK_READS_HIGH = True`.

STEP 4

Keypad and hook to the Pico

30 minutes

1. The board prints the keypad map beside its ribbon: **C3, C2, C1, R4, R3, R2, R1**. Put a marker dot on the **R1** wire, then cut the ribbon close to the board.
2. Turn the Pico **underside up** (the side where the pins stick out and the pin names are printed), USB at the top. Pins 1–20 run down the right edge.
3. Connect each wire as in the map. For the prototype, solder each wire to half a female-to-female jumper and push it onto the pin; hot-glue the rows once it all works.
4. **Volume:** the front panel's 4 wires (ringer + volume sliders) go to GP16, GP17, GP18, GP19 (pins 21, 22, 24, 25), in any order. Then move the VOLUME slider with the Pico on the laptop and tell Claude what it reports; it fills in the setting (full manual, Part 9).



The columns go in reverse on purpose (C1 → GP6, C3 → GP4), so the Pico's code works unchanged.

> **WATCH FOR BRIDGES**
>
> Pins are only 2.54 mm (0.1 in) apart. After each joint, touch the meter to the neighbouring pins: they must not beep together.

STEP 5

Load the Pico

10 minutes

1. Hold the white **BOOTSEL** button, plug the Pico into a laptop, then let go. A drive called **RPI-RP2** appears.
2. From circuitpython.org, download CircuitPython for **Pico W** (or **Pico** if the board doesn't say "Pico W") and drag the `.uf2` file onto RPI-RP2. It comes back as **CIRCUITPY**.
3. Copy everything in `booth/pico/` onto CIRCUITPY: `code.py`, `keymonitor.py`, `keyfinder.py` and the `lib` folder.

CHECK

Lift the handset: the Pico's small LED lights. Open a text editor and press each key: it types its digit. The settings already match the MS-303 wired as in step 4. A wrong digit? Tell Claude what each key typed; it's a one-line fix.

STEP 6

Close up the phone

15 minutes

1. Once everything works, slide the Pico into the clear 1" heat-shrink with the USB end open, and shrink it with the heat gun on low. This stops it shorting on the metal back plate.
2. Stick the Pico and the Sabrent inside with adhesive mounts, away from the metal. Add a screw through each mount where the plastic allows.
3. Zip-tie the wires next to the Pico, so a tug pulls on the ties and never on a solder joint.
4. Run both USB cables out through the old **LINE** jack hole, anchor them inside with a zip tie, and close the phone.

STEP 7

Set up the Pi

30 minutes, mostly waiting

1. Assemble the CanaKit: heatsink on the chip, fan plugged into the socket marked **FAN**, case closed.
2. Plug in the screen (micro-HDMI to mini-HDMI), a keyboard and Ethernet. Plug in power last.
3. When the desktop appears, open Terminal and type these lines, pressing Enter after each:

```
git clone -b claude/landline-demo-booth-setup-05lxch https://github.com/lizzieoh/hel
lo-world.git
cd hello-world/booth/native
./setup-native.sh 1
sudo reboot
```

The `1` is the booth number: use 2, 3 or 4 on the other Pis. Paste the fal key when it asks (the letters don't show; that's normal). The self-test at the end should say every test passed.

STEP 8

Connect everything

10 minutes

FROM	PLUG INTO
Phone: Sabrent USB cable	Pi USB port
Phone: Pico USB cable	Pi USB port
Printer USB cable	Pi USB port
Screen power	Its own USB-C charger, never the Pi
Pi, screen and printer chargers	Power strip

STEP 9

Test call

15 minutes

- [ ] Lift the handset: you hear a ring, a click, the greeting, then a beep
- [ ] Say a prompt: "please hold" and music play, then the video appears, muted
- [ ] Press `2`: sound on. Press `3`: a new take. Press `0`: the receipt prints and cuts.
- [ ] Hang up: the screen loops recent videos
- [ ] Unplug the screen's power and plug it back in: it switches itself back on
- [ ] Pull the Ethernet while on hold: you hear a busy signal. Plug it back in: the next call works.
- [ ] 20 calls in a row with no problems: this booth is done

SHOW DAY

Quick fixes

PROBLEM	FIX
Something's stuck	Hold `*` for 3 seconds. The app restarts in a few seconds.
Time to power off	Hold `#` for 5 seconds, wait for the Pi's green light to stop blinking, then unplug.
Red dot on the screen	The network is down. Check the router.
"Printer offline"	Replug the printer's USB cable, then press `0` again.
A booth is dead	Swap in the spare. Any booth's memory card works in any Pi.